Computer Network Design Project

Two-Floor Corporate Office Building

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Chapter 1

Wiring

The building is a two-floor corporate office complex, where the **ground floor** is served by two distribution frames: the **Main Distribution Frame (MDF)** and **Intermediate Distribution Frame IDF1**. The **first floor** is served by **IDF2** and **IDF3**. Rooms are numbered with the floor prefix: `1.x` for the ground floor and `2.x` for the first floor. Bathrooms and utility rooms (1.3, 1.9, 1.10, 1.20, 1.21, 1.28, 1.34, 1.35 and their first-floor counterparts) are excluded from network cabling.

Room types used throughout the document are defined in Table 1.1.

Table 1.1: Room Type Definitions

Type	**Description**	**Area ($\mathbf{m^2}$)**	**Dimensions (m)**
A	Standard Office	55.25	6.5×8.5
B	Reception / Lobby	55.25	8.5×6.5
C	Individual Office	46.75	5.5×8.5
D	Large Conference Room	97.75	8.5×11.5
E	Department Office	74.75	6.5×11.5
F	Open-Plan Office Space	330.00	20×16.5
G	Large Meeting Room	200.00	10×20
H	Computer Lab	90.00	9×10
I	Seminar Room	58.50	6.5×9
J	Small Office	42.75	4.5×9.5
K	Collaborative Office	68.75	5.5×12.5
L	Multi-Purpose Room	70.00	5×14

1.1 Double Outlets

1.1.1 Calculation Methodology

Double RJ-45 Cat 6 outlets are installed in every networked room. The minimum density follows the standard of **one double outlet per 10 $\mathbf{m^2}$** of usable floor area, rounded up, with a minimum of one double outlet per planned workstation position.

Each double outlet provides **two simple (single) RJ-45 sockets**.

$$N_{\text{double}} = \left\lceil \frac{A_{\text{room}}}{10} \right\rceil, \qquad N_{\text{simple}} = 2 \times N_{\text{double}} \tag{1.1}$$

1.1.2 Explicit Calculation Example – Room 1.1 (Type A)

- **Room:** 1.1, Type A, Standard Office
- **Dimensions:** 6.5 m × 8.5 m
- **Floor area:** $A = 6.5 \times 8.5 = 55.25\,\text{m}^2$
- **Distribution frame:** MDF

Applying Equation 1.1:

$$N_{\text{double}} = \left\lceil \frac{55.25}{10} \right\rceil = \lceil 5.525 \rceil = \mathbf{6} \text{ double outlets}$$

$$N_{\text{simple}} = 2 \times 6 = \mathbf{12} \text{ simple outlets}$$

1.1.3 Ground Floor Outlet Summary

Table 1.2: Ground Floor – MDF Zone Outlets

Room	Type	L (m)	W (m)	Area ($\mathbf{m^2}$)	Double	Simple
1.1	A	6.5	8.5	55.25	6	12
1.2	B	8.5	6.5	55.25	6	12
1.4	C	5.5	8.5	46.75	5	10
1.5	C	5.5	8.5	46.75	5	10
1.6	C	5.5	8.5	46.75	5	10
1.7	C	5.5	8.5	46.75	5	10
1.8	C	5.5	8.5	46.75	5	10
1.11	D	8.5	11.5	97.75	10	20
1.12	F	20.0	16.5	330.00	33	66
1.13	E	6.5	11.5	74.75	8	16
1.14	E	6.5	11.5	74.75	8	16
1.15	E	6.5	11.5	74.75	8	16
1.16	E	6.5	11.5	74.75	8	16
1.17	E	6.5	11.5	74.75	8	16
1.22	G	10.0	20.0	200.00	20	40
			MDF Subtotal		**140**	**280**

Table 1.3: Ground Floor – IDF1 Zone Outlets

Room	Type	L (m)	W (m)	Area (m^2)	Double	Simple
1.18	E	6.5	11.5	74.75	8	16
1.19	E	6.5	11.5	74.75	8	16
1.23	G	10	20.0	200.00	20	40
1.24	H	9	10.0	90.00	9	18
1.25	H	9	10.0	90.00	9	18
1.26	I	6.5	9.0	58.50	6	12
1.27	I	6.5	9.0	58.50	6	12
1.29	J	4.5	9.5	42.75	5	10
1.30	J	4.5	9.5	42.75	5	10
1.31	J	4.5	9.5	42.75	5	10
1.32	J	4.5	9.5	42.75	5	10
1.33	J	4.5	9.5	42.75	5	10
1.36	K	5.5	12.5	68.75	7	14
1.37	K	5.5	12.5	68.75	7	14
1.38	K	5.5	12.5	68.75	7	14
1.39	K	5.5	12.5	68.75	7	14
1.40	K	5.5	12.5	68.75	7	14
1.41	K	5.5	12.5	68.75	7	14
			IDF1 Subtotal		**133**	**266**

Ground Floor Total: 273 double outlets = 546 simple outlets

1.1.4 First Floor Outlet Summary

The first floor mirrors the ground floor in the IDF2 zone (same room types as the MDF zone). The IDF3 zone differs slightly: no H-type computer labs, instead, a combined L-type room is present.

Table 1.4: First Floor – IDF2 Zone Outlets (mirrors MDF zone)

Room	Type	L (m)	W (m)	Area (m^2)	Double	Simple
2.1	A	6.5	8.5	55.25	6	12
2.2	B	8.5	6.5	55.25	6	12
2.4	C	5.5	8.5	46.75	5	10
2.5	C	5.5	8.5	46.75	5	10
2.6	C	5.5	8.5	46.75	5	10
2.7	C	5.5	8.5	46.75	5	10
2.8	C	5.5	8.5	46.75	5	10
2.11	D	8.5	11.5	97.75	10	20
2.12	F	20.0	16.5	330.00	33	66
2.13	E	6.5	11.5	74.75	8	16

Table 1.4 *(continued)*

Room	Type	L (m)	W (m)	Area (m^2)	Double	Simple
2.14	E	6.5	11.5	74.75	8	16
2.15	E	6.5	11.5	74.75	8	16
2.16	E	6.5	11.5	74.75	8	16
2.17	E	6.5	11.5	74.75	8	16
2.22	G	10.0	20.0	200.00	20	40
				IDF2 Subtotal	**140**	**280**

Table 1.5: First Floor – IDF3 Zone Outlets

Room	Type	L (m)	W (m)	Area (m^2)	Double	Simple
2.18	E	6.5	11.5	74.75	8	16
2.19	E	6.5	11.5	74.75	8	16
2.23	G	10	20.0	200.00	20	40
2.24/25	L	5.0	14.0	70.00	7	14
2.26	I	9.0	6.5	58.50	6	12
2.27	I	6.5	9.0	58.50	6	12
2.29	J	4.5	9.5	42.75	5	10
2.30	J	4.5	9.5	42.75	5	10
2.31	J	4.5	9.5	42.75	5	10
2.32	J	4.5	9.5	42.75	5	10
2.33	J	4.5	9.5	42.75	5	10
2.36	K	5.5	12.5	68.75	7	14
2.37	K	5.5	12.5	68.75	7	14
2.38	K	5.5	12.5	68.75	7	14
2.39	K	5.5	12.5	68.75	7	14
2.40	K	5.5	12.5	68.75	7	14
2.41	K	5.5	12.5	68.75	7	14
				IDF3 Subtotal	**122**	**244**

First Floor Total: 262 double outlets = 524 simple outlets

Table 1.6: Building Outlet Grand Total

Distribution Frame	Double Outlets	Simple Outlets	Cable (m)
MDF	140	280	16,980.16
IDF1	133	266	12,258.04
IDF2	140	280	16,980.16
IDF3	122	244	10,899.02
Total	**535**	**1,070**	**57,117.38**

1.2 UTP Cable Quantity Calculation

1.2.1 Calculation Methodology

For each room, the total UTP cable needed equals the number of simple outlets multiplied by the cable length per outlet. The cable length per outlet accounts for:

- the **horizontal distance** from the distribution frame to the room (d_{DF});
- the **average cable span inside the room** to reach each outlet (l_{avg});
- a **technical reserve** of 6 m per cable (3 m at the outlet side + 3 m at the patch-panel side).

$$L_{\mathrm{room}} = N_{\mathrm{simple}} \times \left(d_{\mathrm{DF}} + l_{\mathrm{avg}} + 6\right) \tag{1.2}$$

1.2.2 Detailed Example – Room 1.1

- Simple outlets: $N_{\mathrm{simple}} = 12$
- Distance from room to MDF: $d_{\mathrm{DF}} = 9.4\,\mathrm{m}$
- Average cable run inside room: $l_{\mathrm{avg}} = 14\,\mathrm{m}$
- Technical reserve: $r = 6\,\mathrm{m}$

$$L_{\mathrm{room\ 1.1}} = 12 \times (9.4 + 14 + 6) = 12 \times 29.4 = \mathbf{352.8}\,\mathrm{m}$$

A summary of all rooms and their cable contributions by DF zone is given in Table 1.7.

Table 1.7: UTP Cable Length per Room – MDF Zone (Ground Floor)

Room	Type	d_{DF} (m)	l_{avg} (m)	Simple	Cable (m)
1.1	A	9.40	14	12	352.80
1.2	B	16.11	5	12	325.32
1.4	C	21.82	11	10	388.20
1.5	C	27.82	11	10	448.20
1.6	C	33.53	11	10	505.30
1.7	C	39.53	11	10	565.30
1.8	C	45.24	11	10	622.40
1.11	D	0.50	11	20	350.00
1.12	F	5.00	22	66	2,178.00
1.13	E	84.64	17	16	1,722.24
1.14	E	91.78	17	16	1,836.48
1.15	E	98.78	17	16	1,948.48
1.16	E	105.92	17	16	2,062.72
1.17	E	112.92	17	16	2,174.72
1.22	G	17.50	14	40	1,500.00
			MDF Subtotal	**280**	**16,980.16**

1.2.3 Total UTP Cable Quantities

The complete building cable totals are summarised in Table 1.8. The average calculated cable span across all links is approximately 53.4 m, confirming the design stays well within the Cat 6 maximum segment length of 100 m.

Table 1.8: UTP Cable Total Summary

Parameter	**Value**
Total cable	57,117.38 m
Technical reserve	10%
Total with reserve	62,829 m
Average cable span	≈ 53.4 m

1.3 UTP Cable Boxes

UTP Cat 6 cable is procured in 305 m boxes. The number of boxes required is:

$$N_{\text{boxes}} = \left\lceil \frac{L_{\text{total, with reserve}}}{305} \right\rceil = \left\lceil \frac{62{,}829}{305} \right\rceil = \lceil 205.8 \rceil = \mathbf{207} \text{ boxes} \tag{1.3}$$

Equivalently in 2 km reels:

$$N_{\text{reels}} = \left\lceil \frac{62{,}829}{2{,}000} \right\rceil = \lceil 31.4 \rceil \approx \mathbf{29} \text{ reels (2 km each)}$$

The two procurement options are summarised in Table 1.9.

Table 1.9: UTP Cable Procurement Options

Format	**Length/unit**	**Quantity**	**Total length (m)**
Standard box (305 m)	305 m	207	63,135
Large reel (2 000 m)	2,000 m	29	58,000

1.4 Fiber Optic Cable Calculation

1.4.1 Backbone Topology

The four distribution frames are interconnected by OM4 multimode fiber optic (50/125 µm) in a **redundant ring/mesh topology**, providing multiple paths between any two DFs. The node labelling used in the cable-planning spreadsheet is:

- **A** = MDF **B** = IDF1 **C** = IDF2 **D** = IDF3
- suffix **0** = primary port; suffix **1** = secondary (redundant) port

1.4.2 Detailed Fiber Link Calculation

Table 1.10: Fiber Optic Backbone Links

Source	**Target**	**Length**	**Description**
A0	B0	35 m	MDF → IDF1, ground-floor backbone
B1	A1	85 m	IDF1 → MDF, redundant inter-floor path
C0	D0	35 m	IDF2 → IDF3, first-floor backbone
D1	C1	85 m	IDF3 → IDF2, redundant inter-floor path
A1	C1	5 m	MDF ↔ IDF2, adjacent riser cross-connect
D0	B0	5 m	IDF3 ↔ IDF1, adjacent riser cross-connect
A0	C0	5 m	MDF ↔ IDF2, primary riser cross-connect
D1	B1	5 m	IDF3 ↔ IDF1, primary riser cross-connect
Total (raw)		**260 m**	
+ 15% procurement reserve		**299 m**	rounded to 300 m

Detailed example: Link A0 → B0:

$$L_{\text{raw}} = 35\,\text{m} \quad \text{(measured in floor plan)}$$
$$L_{\text{reserve}} = 35 \times 1.15 = 40.25\,\text{m} \approx 41\,\text{m}$$

Duplex OM4 cable is used, so both transmit and receive fibers are contained in a single cable run. The total cable procurement for the entire backbone is:

$$L_{\text{FO, total}} = 260\,\text{m} \times 1.15 \approx \mathbf{300}\,\text{m} \quad \text{(one 300 m reel of duplex OM4)}$$

Each distribution frame is equipped with one **PP-72X100G-MMF** fiber patch panel (72 LC duplex ports, OM4 compatible), accommodating all backbone connections and future expansion.

Chapter 2

VLANs and IP Address Assignment

2.1 IP Addressing Scheme

The network uses a **private address space** under the block `10.10.0.0/16`, NAT-ed to the public IP address `203.0.113.0`. VLANs are segmented to isolate traffic types and departments. The **Management VLAN** is protected from all other VLANs by firewall policy.

Total network capacity planning:

- Total workstations: 928
- Total IP phones: 67
- Total wireless clients (AP capacity): up to 1,000
- Estimated peak network traffic: **32 Gbps**

2.2 VLAN Assignment Table

Table 2.1: VLAN and IP Address Plan

VLAN	Name	Network	Gateway	Mask	Hosts
10	Management	10.1.10.0	10.1.10.1	/24	254
20	IP_PhonesData	10.1.19.0	10.1.19.1	/24	254
20	IP_Phones	10.1.20.0	10.1.20.1	/24	254
32	WLESS	10.1.32.0	10.1.32.1	/22	1022
40	HR	10.1.40.0	10.1.40.1	/24	254
50	Finance	10.1.50.0	10.1.50.1	/24	254
			Total usable IP addresses		**1526**

Management VLAN security note: VLAN 10 (Management) is accessible exclusively from designated management hosts. Access from all other VLANs to VLAN 10 is **blocked** at the L3 switch and enforced by ACL on the Cisco Catalyst 9410R supervisors. Remote access to all network devices (SSH) is available only through VLAN 10.

Chapter 3

Equipment Selection

For each equipment category, two options are compared. **Option 1 (Cisco)** was selected as the preferred choice, with a justification given in Section 3.10.

3.1 L2/L3 Modular Switch Chassis

Table 3.1: Switch Chassis Comparison

Specification	**Option 1: Cisco Catalyst 9410R**	**Option 2: HPE Aruba CX 6410 v2**
Chassis slots	10 line-card slots	6 line-card slots
Max access ports	480× 1 GE or 48× mGig	480× 1 GE PoE
Backplane	3.2 Tbps	1.6 Tbps
OS	Cisco IOS-XE (DNAC ready)	AOS-CX
L3 routing	Full (OSPF, BGP, EIGRP, VRF)	Full (OSPF, BGP)
Redundant supervisors	Yes (dual SUP slots)	Yes
PoE budget max	8×3 200 W PSU	4×3 600 W PSU
Unit price	$6,100.82	$26,334.70

3.2 Supervisor / L3 Module

Table 3.2: Supervisor Module Comparison

Specification	**Option 1: Cisco C9400-SUP-1XL**	**Option 2: HPE CX 6400 Mgmt Module**
Uplink ports	2×40 GE + 2×25 GE	2×100 GE QSFP28
Forwarding rate	480 Mpps	400 Mpps
Power consumption	350 W	200 W
HA features	In-Service Software Upgrade (ISSU)	ISSU
Management	NetConf, RestConf, YANG	REST API
Unit price	$7,123.14	$14,125.61

3.3 48-Port Access Line Card (Data / IP Phone)

Table 3.3: 48-Port PoE Line Card Comparison

Specification	**Option 1: Cisco C9400-LC-48P**	**Option 2: HPE CX 6400 R0X39C**
Port count	48× 1 GE RJ-45	48× 1 GE RJ-45
PoE standard	802.3at PoE+ (30 W/port)	802.3at PoE+
Max PoE budget	2,880 W	740 W
Throughput per card	96 Gbps	96 Gbps
Unit price	$4,959.83	$5,350.00

3.4 Fiber Optic Line Card (min 40Gbps ports)

Table 3.4: FO Line Card Comparison (min 40 Gbps)

Specification	**Option 1: Cisco C9400-LC-12QC**	**Option 2: HPE CX 6400 R0X45C**
Port count	12× 40 GE QSFP+	8× 100 GE QSFP28
Min port speed	40 Gbps (meets requirement)	100 Gbps
Total bandwidth	480 Gbps	800 Gbps
FO module (Cisco)	QSFP-40G-SR4, $343.64/port	HP JL310A, $1,295/port
OM4 reach	150 m	100 m
Line card price	$17,336.86	$47,650.00

3.5 Multi-Gigabit AP Line Card

Table 3.5: AP Line Card Comparison

Specification	**Option 1: Cisco C9400-LC-48UX**	**Option 2: HPE CX 6400 S1T83A**
Port count	48× mGig (1/2.5/5/10 GE)	24× 2.5 GE SFP+
PoE standard	802.3bt PoE++ (90 W/port)	802.3at PoE+
Max PoE budget	2,880 W	720 W
Ideal use	WiFi 7 APs, high-density	APs up to WiFi 6E
Unit price	$5,015.78	$15,545.00

3.6 Firewall

Table 3.6: Firewall Comparison

Specification	**Option 1: Cisco Secure Firewall 4245**	**Option 2: Palo Alto PA-3440**
FW throughput	70 Gbps	20 Gbps
IPS throughput	50 Gbps	12 Gbps
TLS/SSL inspection	Yes (35 Gbps)	Yes (9 Gbps)
Interfaces	2×100 GE + 16×10 GE	4×10 GE + 12×1 GE
HA support	Active/Active, Active/Standby	Active/Passive
Threat intelligence	Cisco Talos	PAN Threat Prevention
Unit price	$152,257.81	$68,179.32

3.7 WLAN Controller

Table 3.7: WLAN Controller Comparison

Specification	**Option 1: Cisco CW9800L**	**Option 2: HPE Aruba Central**
Form factor	Physical appliance (1U)	Cloud SaaS
AP capacity	1000 APs	Unlimited (subscription)
Throughput	160Gbps	Cloud-based
WiFi 7 (802.11be)	Yes	Yes
On-premises operation	Yes (no Internet required)	Requires Internet
Price model	One-time $3,686.60	Annual subscription

3.8 Access Points (min IEEE 802.11be / WiFi 7)

Table 3.8: Access Point Comparison

Specification	**Option 1: Cisco CW9178I**	**Option 2: HPE Aruba AP-765**
WiFi standard	802.11be (WiFi 7)	802.11be (WiFi 7)
Frequency bands	2.4 / 5 / 6 GHz (tri-band)	2.4 / 5 / 6 GHz (tri-band)
Max throughput	6.5 Gbps	5.76 Gbps
MIMO streams	4×4 (6 GHz), 4×4 (5 GHz)	4×4 (6 GHz)
PoE requirement	802.3bt PoE++ (60 W)	802.3bt PoE++ (30 W)
Uplink port	2.5 GE mGig	2.5 GE mGig
Unit price	$5,455.89	$1,475.37

3.9 IP Phones

Table 3.9: IP Phone Comparison

Specification	**Option 1: Cisco IP Phone 8861**	**Option 2: HP 82M93AA (HP 4110)**
Display	5 in colour VGA (800×480)	2.3 in colour
Wi-Fi	802.11ac dual-band	802.11ac
Bluetooth	4.2 (headset support)	No
USB	2× USB (PC/charging)	No
Lines	5 SIP lines	2 SIP lines
PoE standard	802.3at PoE+	802.3af PoE
Codec support	G.711, G.722, G.729, Opus	G.711, G.729
Unit price	$366.20	$194.99

3.10 Choice Justification

Technical Arguments for Cisco

1. **Unified ecosystem:** Cisco Catalyst 9000 switches, CW9178I APs, CW9800L controller, and Secure Firewall 4245 are all managed from a single *Catalyst Center* dashboard, reducing operational complexity.

2. **High-availability design:** The 9410R supports dual redundant supervisors, per-slot power-supply redundancy, and In-Service Software Upgrade (ISSU), enabling zero-downtime maintenance.

3. **Scalability:** The modular 9410R chassis allows future line-card additions without replacing the chassis investment.

Monetary Arguments for Cisco

The total cost for the Cisco option (excluding Firewall, UPS, and cooling) is **$710,978.24**, compared to **$791,727.64** for HPE Aruba, a saving of approximately **$80,749.40 (10.2%)**.

Chapter 4

Environment

4.1 Rack Size Calculation and Choice

4.1.1 Rack Layout per Distribution Frame

Each DF is housed in a single **44U Open-Frame Server Rack**. The rack layout is built from the bottom (UPS) upward, with ventilation clearance rows between heat-generating equipment groups.

Table 4.1: MDF Rack Layout (44U total)

Space (U)	Position	Equipment	Model	Power (W)
1	Top	OF Patch Panel	PP-72X100G-MMF	–
13		L2/L3 Switch Chassis	Cisco Catalyst 9410R	–
		Supervisor ×2	C9400-SUP-1XL	350×2
		OF Line Card	C9400-LC-12QC	280
		WS LC ×5	C9400-LC-48P	65×5
		IP Phone LC ×1	C9400-LC-48P	65
		AP LC ×1	C9400-LC-48UX	1,630
		Fan Tray	C9410-FAN	600
		PSU ×3	PWR-C4-3200WAC	3,200×3
14		UTP Patch Panel ×7	CPPKL6ATG48WBL	–
2		Ventilation clearance	–	–
1		Firewall	Cisco FW 4245	1,380
2		Ventilation clearance	–	–
12	Bottom	UPS ×2	APC SRT10KXLT	–
44	**Total**			

Table 4.2: IDF1 / IDF2 / IDF3 Rack Layout (39U used of 44U)

Space (U)	Equipment	Model	Power (W)
1	OF Patch Panel	PP-72X100G-MMF	–
13	L2/L3 Switch Chassis (same config as MDF, no Firewall)	Cisco Catalyst 9410R	–
14	UTP Patch Panel ×7	CPPKL6ATG48WBL	–
12	UPS ×2	APC SRT10KXLT	–
40	**Total used**	5U expansion headroom	

4.1.2 Rack Options

Table 4.3: Rack Comparison

Specification	Option 1: APC NetShelter SX 44U	Option 2: Rittal TS IT 44U
Height	44U (2,033 mm)	44U (2,055 mm)
Type	Open-frame, 4-post	Enclosed, 4-post
Load capacity	1,360 kg	1,500 kg
Rail depth	Adjustable (480–900 mm)	Fixed 800/1,000/1,200 mm
Airflow	Excellent (open frame)	Managed (perforated doors)
Price per unit (USD)	≈$800	≈$1,200
Quantity	**4**	**4**
Total	**$3,200**	**$4,800**

Choice: APC NetShelter SX 44U open-frame rack. The open-frame design maximises natural airflow and reduces cooling load, is $400 cheaper per unit, and provides easier front-rear cable access during maintenance.

4.2 UPS Power Calculation and Choice

4.2.1 Power Requirement per Distribution Frame

The required UPS power is the sum of all powered equipment in the rack, with a 30% safety margin per project specification. UPS units only supply network and server equipment; cooling units are on a separate AC circuit.

Table 4.4: UPS Power Calculation per DF

Parameter	MDF	IDF1	IDF2	IDF3
Chassis power (W)	4,030	4,030	4,030	4,030
Firewall (W)	1,380	–	–	–
Total heat (BTU/h)	17,013	13,553	13,553	13,553
Required UPS power (W)	11,115	9,600	9,600	9,600
Available UPS power (W)	20,000	20,000	20,000	20,000
Safety margin	80%	108%	108%	108%

Two **APC SRT10KXLT** units per DF give 20 kW available, which is 80–108% above the required load.

4.2.2 UPS Options

Table 4.5: UPS Comparison

Specification	**Option 1: APC SRT10KXLT**	**Option 2: Eaton 9PX 10kVA**
Rated power	10,000 VA / 10,000 W	10,000 VA / 9,000 W
Technology	Online double-conversion	Online double-conversion
Form factor	Tower / 6U rack-convertible	6U rack
Transfer time	0 ms (always online)	0 ms
Runtime (50% load)	≈15 min (expandable)	≈15 min
Management	SNMP / Web (card option)	Network-MS card
Unit price (USD)	≈\$4,500	≈\$4,200
Qty (2/DF ×4 DF)	**8**	**8**
Total (USD)	**≈\$36,000**	≈\$33,600

Choice: APC SRT10KXLT. Wider availability, established service network, and seamless integration with APC EcoStruxure infrastructure management platform. The 10 kW rating provides a larger power buffer, which is preferred for the MDF rack housing the firewall.

4.3 Cooling Capacity Calculation and Choice

4.3.1 Heat Dissipation per Distribution Frame

Total heat dissipated must be removed by the cooling system. A minimum 30% additional cooling capacity is required above the calculated maximum.

Table 4.6: Cooling Requirement per DF

Parameter	**MDF**	**IDF1**	**IDF2**	**IDF3**
Total heat (BTU/h)	17,013	13,553	13,553	13,553
Required (+30%) (BTU/h)	22,117	17,619	17,619	17,619
Installed cooling (BTU/h)	38,000	38,000	38,000	38,000
Actual safety margin	123%	180%	180%	180%
AC units per DF	2	2	2	2

Two **Daikin Sky Air ADEAS60A** units per DF provide 38,000 BTU/h.

4.3.2 Cooling Options

Table 4.7: Cooling Unit Comparison

Specification	**Option 1: Daikin ADEAS60A**	**Option 2: Mitsubishi PEAD-M60JA**
Type	Ducted air conditioner	Ceiling-concealed ducted
Cooling capacity	6.0 kW / 19,070 BTU/h	6.0 kW / 20,480 BTU/h
Heating capacity	7.0 kW	7.0 kW
Energy rating	A++ (SEER 7.4)	A+ (SEER 6.8)
Sound level (indoor)	28 dB(A)	30 dB(A)
Refrigerant	R-32	R-32
Electric input	1,600 W (cooling)	1,700 W (cooling)
Unit price (USD)	≈$3,500	≈$3,200
Total (8 units)	**≈$28,000**	≈$25,600

Choice: Daikin ADEAS60A. Higher energy efficiency reduces long-term operating costs, quieter operation, and Daikin's VRV/VRF technology allows centralised monitoring of all 8 units.

Chapter 5

Active Equipment List

TODO

Chapter 6

Material List

TODO

Chapter 7

Network Drawings

7.1 Floor Plan – Ground Floor

Figure 7.1: Ground floor plan showing room layout, MDF and IDF1 locations, and main cable tray paths.

Key features:

- MDF room located adjacent to Room 1.11
- IDF1 located at the far wing, adjacent to Room 1.23
- Main cable tray runs along the corridor spine
- 22 networked rooms, bathrooms (1.3, 1.9, 1.10, 1.20, 1.21, 1.28, 1.34, 1.35) excluded

7.2 Floor Plan – First Floor

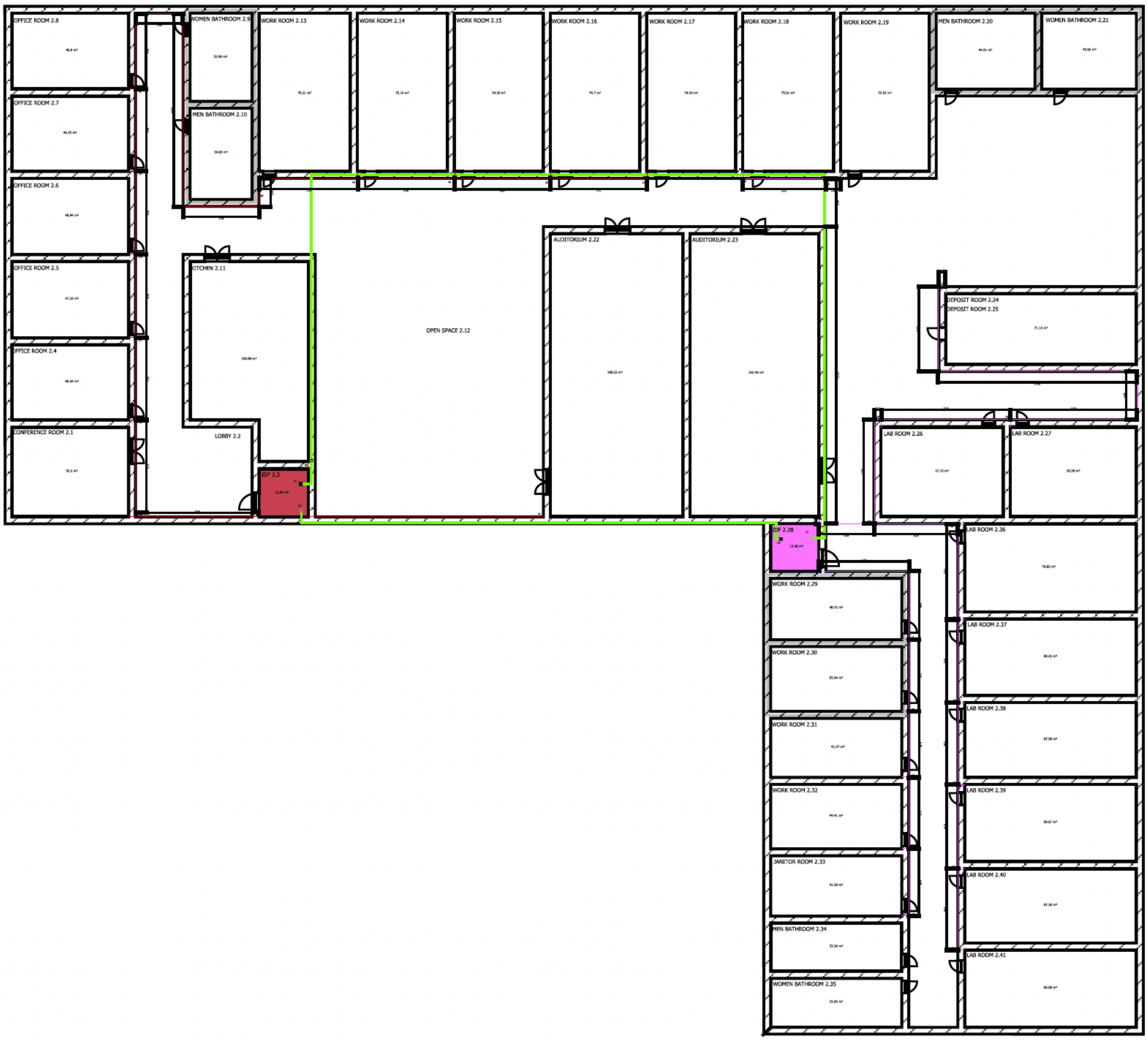

Figure 7.2: First floor plan showing room layout, IDF2 and IDF3 locations, and main cable tray paths.

Key features:

- IDF2 mirrors the MDF position on the floor above (adjacent to Room 2.11)
- IDF3 mirrors the IDF1 position on the floor above (adjacent to Room 2.23)
- Riser ducts in the building's vertical shaft connect floor DFs via FO backbone
- First floor has an additional L-type multi-purpose room (2.24/25) instead of H-type labs

7.3 Cable Path Plans

Detailed cable routing drawings with raceway paths, outlet positions, and DF locations are provided above...

Also maybe here we can include the planning of each individual room...

7.4 Network Logical Topology

```
  Internet
     |
 [Cisco FW 4245] -- VLAN 10 (Management)
     |
 [MDF -- Cisco 9410R L3 switch / firewall uplink]
  /  |  \  \
 FO  FO  FO  FO  (40GE backbone -- ring/mesh)
 /    |    \   \
IDF1  IDF2  IDF3   (via 35m + 85m + 5m runs)
 |     |     |
L2    L2    L2   (48-port PoE line cards)
 |     |     |
WS, Phones, APs per floor zone
```

7.5 Rack Diagrams

Rack layout diagrams for all four distribution frames are described in Chapter 4 (Tables 4.1 and 4.2).

TODO... maybe add something more visual

Chapter 8

References

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11. SweetHome3D Floor Planning Tool (used for building layout):
https://www.sweethome3d.com/